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SYSTEMATIC REVIEW

Diagnosis melanoma with artificial intelligence systems: A meta-analysis study and systematic review

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Abstract

Background: One of the most promising and rapidly advancing research areas in recent years is using dermoscopic images for automatic diagnosis with artificial intelligence and machine learning methods.

Objectives: This study aimed to synthesize the existing studies for the clinical use of applications made with artificial intelligence methods and to summarize the predictive performance of deep learning and hybrid models-based algorithms in all these studies with a large-scale meta-analysis.

Method: The literature review was conducted between January 2006 and May 2024, and meta-analysis data were created by scanning the Web of Science (WOS), Scopus and MEDLINE databases. This study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) checklist.

Results: A total of 2722 articles were evaluated. Data from 78 diagnostic tests from 39 primary studies meeting the inclusion and exclusion criteria were assessed. The pooled SROC overall model AUC was 0.96 [95% CI: 0.94–0.98], sensitivity was 0.89 [95% CI: 0.85–0.91] and specificity was 0.92 [95% CI: 0.90–0.94]. In the subgroup analyses, the pooled AUC was 0.98 [95% CI: 0.96–0.99] for HYBRID models.

Conclusions: Recent studies have suggested that artificial intelligence algorithms and machine learning methods should be used extensively in medicine to assist physicians, especially in diagnosing melanoma. The ability of HYBRID model algorithms to predict diseases is promising. In particular, the performance of HYBRID models was found to be high. This information can assist clinicians in interpreting the most appropriate algorithms for diagnosing melanoma.

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INTRODUCTION

Skin cancer is a prevalent and potentially fatal condition affecting millions globally.[1] Recent studies have highlighted its rapid increase, with mortality rates expected to rise in the coming decade. According to WHO (2017),[2] one in three cancer diagnoses is skin cancer, with approximately 132,000 cases of melanoma annually. Early diagnosis is critical, as it significantly improves treatment outcomes and reduces mortality.[3]

In diagnosing melanoma, experienced dermatologists visually assess suspicious lesions, often supplementing this with dermoscopy, which provides microscopic classification of lesions. Dermoscopy, a non-invasive in vivo diagnostic technique, is essential in detecting benign and malignant neoplasms and gives crucial information in diagnosing various skin conditions.[4] The final stage of diagnosis requires a biopsy. However, these processes are time-consuming, and limited access to skilled professionals in developing countries poses a barrier. The accuracy of diagnosis also varies according to practitioners' skills and experience, with few qualified dermatologists available in public health sectors globally.[1,5] Consequently, efforts to enhance diagnostic accuracy have driven the development of innovative approaches to melanoma diagnosis.[6]

Recent research has increasingly focused on improving early melanoma diagnosis, with particular attention to machine-learning (ML) and artificial intelligence (AI) applications. AI simulates human behaviour through machines and programs, while ML enables machines to learn from historical data automatically.[7,8]

Dermatology, in particular, stands to benefit significantly from AI, where image-based computer-aided diagnosis (CAD) systems are becoming more common due to their potential in screening and detecting melanoma from dermoscopic images.[9,10] Various subsets of AI, such as ML, deep learning (DL) and artificial neural networks (ANNs), play prominent roles in dermatology.[11] Within ML, support vector machine (SVM) has gained recognition for its high performance in predicting both linear and non-linear relationships.[12] Some studies propose hybrid models that integrate SVM, sparse coding, and DL to enhance melanoma classification accuracy.[13,14]

Convolutional neural networks (CNNs), a specialized form of feedforward neural network, have emerged as a key tool in image classification due to their ability to recognize images regardless of orientation. CNNs have driven the development of powerful architectures like AlexNet, GoogleNet Inception v3, ResNet and DenseNet, which evolved into Deep CNNs (DCNNs) through the addition of hidden layers, enhancing their ability to handle complex image tasks.[13,14] As a result, DCNNs have gained attention in medical imaging, where their capability to support diagnostic processes has proven invaluable.[13]

In recent years, 'multi-network systems', which combine CNNs, DCNNs and neural networks, have been developed to improve diagnostic accuracy. These HYBRID models such as GoogleNet-MobileNet and MobileNet-DenseNet leverage the strengths of multiple DL networks.[15] Researchers are also exploring hybrid variations like GoogleNet-VGG16-ANN and AlexNet-GoogleNet-ANN and AlexNet-GoogleNet-VGG16-Random Forest to further enhance melanoma diagnosis.[16] Such HYBRID models aim to achieve higher diagnostic accuracies by integrating the strengths of different DL networks.

This study aimed to synthesize existing studies on the clinical use of applications of AI methods and to evaluate the performance of the algorithms used in all these studies. In this context, a large-scale meta-analysis is planned to assess collectively and comparatively analyse these algorithms and compare their predictive performance in melanoma diagnosis. We aim to obtain essential findings that will support the decision of dermoscopy specialists regarding melanoma diagnosis.

Why was the study undertaken?

The study explores the integration of AI and machine learning in diagnosing melanoma, focusing on hybrid models that enhance classifier performance through effective feature selection in melanoma detection.

What does this study add?

The study demonstrates that hybrid models improve melanoma diagnosis by optimizing relevant feature extraction, enhancing both the interpretability and accuracy of AI systems.

What are the implications of this study for disease understanding and/or clinical care?

This research underscores the role of feature selection in AI, potentially improving melanoma diagnosis accuracy and clinical decision-making, ultimately benefiting patient outcomes in the diagnosis of melanoma.

MATERIALS AND METHODS

This study was conducted according to the Preferred Reporting Items for Systematic Reviews and the Meta-Analysis (PRISMA) checklist. Ethical approval was not required.

Literature search strategy and study selection

Literature searches were conducted in Web of Science (WOS), Scopus, and MEDLINE databases between January 2006 and May 2024. Search results were extracted from all databases by researchers (G.E.Z, S.I.Y.T, S.Y.I.) and transferred to Covidence (https://www.covidence.org/), an online systematic review tool by a single researcher (E.Ç.G.). Using

the automatic duplication function of Covidence, 315 duplicate publications were identified and filtered from the study sample. After removing duplicate publications, the resulting study set was independently screened by five researchers (S.Y.I., G.E.Z., S.I.Y.T., E.Ç.G. and Ö.F.E.) in five consecutive rounds to distinguish studies that met pre-specified eligibility criteria (depending on inclusion and exclusion criteria). In the initial screening, five researchers (S.Y.I., G.E.Z., S.I.Y.T., E.Ç.G. and Ö.F.E.) independently voted on the titles and abstracts of records obtained from the search via the Covidence portal, with conflicts resolved by consensus. At this stage, two researchers (G.Z. and R.E.) decided whether or not to include studies with conflicts. Four researchers (G.E.Z, S.I.Y.T., A.E and I.E) independently assessed the quality of each study. Inclusion criteria were 'The use of dermatoscopy or dermoscopy methods in diagnosis, using AI algorithms, having melanoma, publishing in English, skin lesion images, classification'. Exclusion criteria were 'unspecified effect size, insufficient data, using other methods rather than dermoscopy in diagnosis, other types of cancer (blood cancer, lung cancer, gastric cancer, stomach cancer, bladder cancer, breast cancer), publishing in languages other than English, COVID-19, DNA microarray data, flow cytometry, genetic algorithm, confocal microscopy'. With AI, search strategies were created by making the most appropriate combinations of keywords to search for studies on skin cancer or melanoma diagnosis, dermatoscopy and dermoscopy. Detailed search strategies are included in the supplementary (Figure S1). Data collection strategies were created in a specific format to analyse the research data.

Data extraction

Two investigators (S.I.Y.T and G.E.Z) independently extracted and reconciled data from eligible studies to ensure accuracy. Authors were systematically contacted for missing data, starting with the corresponding author, followed by the first and last authors. If authors were unreachable, studies were excluded from the meta-analysis but included in the systematic review. Articles or abstracts lacking adequate evaluation data were omitted upon communication with authors. We collectively extracted data from the included articles into a Microsoft Excel spreadsheet. Information criteria: The extracted variables were as follows: (1) title; (2) first author's last name; (3) year of publication; (4) source of data; (5) country of the data used; (6) size of the dataset; (7) analytical models; (8) number of patients. The extracted performance measures were as follows: (9) accuracy; (10) sensitivity; (11) specificity; (12) true-positives (TP); (13) false-positives (FN), (14) true-negatives (TN) and (15) false-negatives (FN).

Quality assessment

The Quality Assessment for Studies of Diagnostic Accuracy II (QUADAS-2) checklist is utilized to assess the quality of diagnostic performance studies.[17] The assessment was performed by five researchers (G.E.Z., S.I.Y.T., A.E., S.Y.I. and I.E.). This checklist, covering four core areas, serves as a quality assessment tool to rate bias risk and research applicability, carefully evaluated during each field study's implementation. Field headings in the checklist: Domain-1: patient selection (risk of bias, concerns regarding applicability); Domain-2: index test(s) (risk of bias, concerns regarding applicability); Domain-3: reference standard (risk of bias, concerns regarding the applicability); and Domain-4: flow and timing (risk of bias). All checklists of QUADAS-2 can be accessed from the website (www.quadas.org). The risk of bias for each domain was determined using the signalling questions in QUADAS-2. Each signalling question was rated as 'no', 'unclear' or 'yes'. The domains were considered a low risk of bias if all the signalling questions were rated 'yes'. However, the domains were considered at a high risk of bias if one of the signalling questions was rated 'no'. Except for the previous two conditions, the domains were considered an unclear risk of bias.

Statistical analysis

AI used a bivariate meta-analysis technique to obtain joint sensitivity, specificity and positive likelihood ratio for melanoma detection. Summary receiver operator characteristic curves (SROC) were generated to explore the pooled diagnostic value of melanoma detection with AI. Fagan's nomogram of positive and negative likelihood ratios (PLR, NLR) was used to assess their clinical application value. We evaluated heterogeneity using two methods (chi-square-based Q statistic Cochran's Q and I^2). A p-value >0.1 and $I^2 < 50\%$ indicate that heterogeneity is not statistically significant, and the fixed-effect model should be chosen. Otherwise, the random-effects model would be more appropriate.[18] Two scenarios were created to compare the predictive performance of the algorithms. In the first scenario, ML, DL and HYBRID models were evaluated, and in the second scenario, CNN, DCNN, HYBRID and SVM algorithms were assessed. Additionally, publication bias was estimated by Egger's test. The Covidence tool was used to evaluate the meta-analysis (https://www.covidence.org/) qualitatively. Quality assessment was made using the QUADAS-2 checklist. All analyses were performed with the 'midas' package in Stata/SE 17.0 (StataCorp LLC, USA) and the 'metaphor' package in R language (Version 2021.09.1 Build 372-©2009–2021 RStudio, PBC). The significance level was accepted as $p < 0.05$.

RESULTS

Study selection and characteristics of eligible studies

The initial search identified 2722 records, of which 315 duplicates were removed using referencing software. A further 1723 studies were removed according to titles and abstracts.

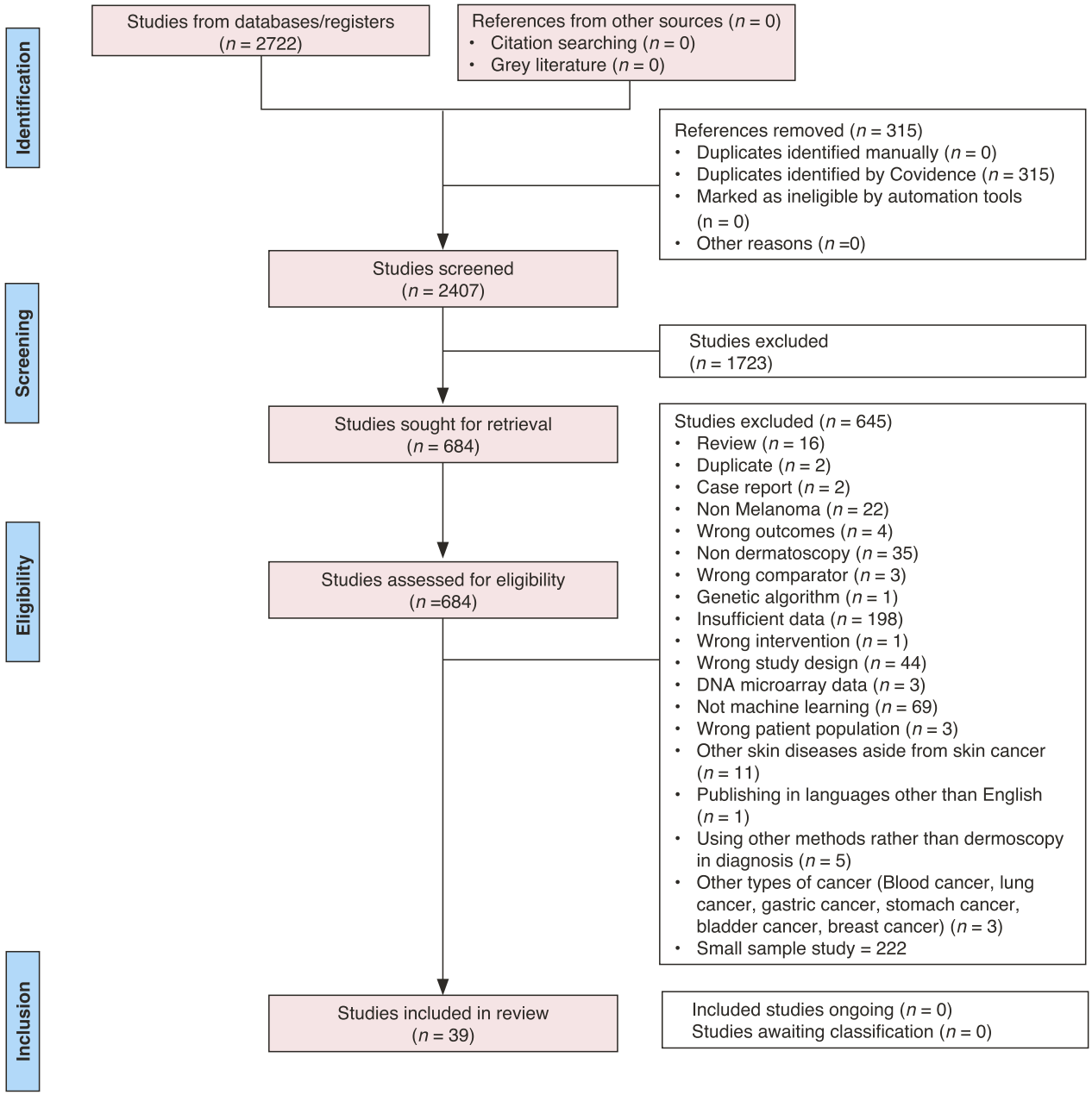


FIGURE 1 PRISMA flow diagram of the study selection process. PRISMA, Preferred Reporting Item for Systematic Reviews and Meta-Analyses.

In total, 684 records on melanoma were included, and after reading 684 studies' full texts, 645 articles were excluded. Eventually, 39 articles were included in the review for the meta-analysis. Due to the exclusion of studies that did not meet the inclusion criteria and the exclusion of studies with small samples from the study sample, the dataset consisted of 78 diagnostic test data from 39 primary studies (Figure 1). Moderator variables used in the study were determined as AI models. The author information of the studies included in the study, years, country, used AI models, AI models scenario-I (DL, HYBRID, ML) and AI models scenario-II (CNN, DCNN, HYBRID, SVM) are given descriptively in Table 1.

Descriptive statistics

In the AI models, Scenario-I included 54 studies using DL models, 18 using HYBRID models and 6 using ML models. In Scenario-II, the studies comprised 14 using CNN models, 40 using DCNN models, 18 using HYBRID models and 6 using SVM models. Totally, 78 AI model studies were analysed from 39 selected studies. Regarding the annual distribution of studies, 3 studies from 2024, 13 studies from 2023, 11 studies from 2022, 4 studies from 2021, 2 studies from 2020 and 6 studies from 2019 were included in the analysis (Table 1).

TABLE 1 Characteristics of included studies.

Authors	Country	Used AI models	AI models (Scenario I)	AI models (Scenario II)
Afza et al. (2019)[19]	Pakistan	SVM	ML	SVM
Hosny (2019)[20]	Egypt	DCNN	DL	DCNN
Kassani et al. (2019)[21]	Kanada	CNN	DL	CNN
Unver (2019)[22]	Türkiye	DCNN	DL	DCNN
Madooei (2019)[23]	Kanada	mi-SVM	ML	SVM
Tang (2019)[24]	China	CNN	DL	CNN
Guo et al. (2020)[25]	Switzerland	DCNN	DL	DCNN
Tiwari (2020)[26]	India	CNN	DL	CNN
Anand (2021)[27]	India	CNN, ResNet18, ResNet50, ResNet101	DL	CNN, DCNN
Banerjee et al. (2021)[28]	India	YOLO, Linear SVM	DL, ML	DCNN, SVM
Bansal et al. (2021)[29]	India	SVM + KNN	HYBRID	HYBRID
Thanh (2021)[30]	USA	CNN	DL	CNN
Alfi et al. (2022)[31]	Bangladesh	SVM	ML	SVM
Aljohani and Turki (2022)[32]	Saudi Arabia	GoogleNet	DL	DCNN
Araujo (2022)[33]	Brazil	U-Net	DL	DCNN
Bassel et al. (2022)[34]	Iraq	SVM	ML	SVM
Damarla (2022)[35]	India	NasNet Large, NasNet Large PSO, NasNet Large ACO	DL	DCNN
Hosny et al. (2022)[36]	Egypt	RDCNN	DL	DCNN
Mahum et al. (2022)[37]	Pakistan	LSTM (CNN)	DL	CNN
Nie et al. (2022)[38]	Sweden	ResNet-50	DL	DCNN
Qureshi et al. (2022)[39]	India	CNN	DL	CNN
Wang et al. (2022)[40]	Singapore	ResNet50	DL	DCNN
Yilmaz et al. (2022)[41]	Türkiye	MobileNetV1, MobileNetV2, NASNetMobile, Xception	DL	DCNN
Akbulut et al. (2023)[42]	Türkiye	CNN	DL	CNN
Alenezi et al. (2023)[15]	Saudi Arabia	DenseNet, MobileNet, GoogleNet, GoogleNet+MobileNet, MobileNet+DenseNet	DL, HYBRID	DCNN, HYBRID
Baig et al. (2023)[43]	Saudi Arabia	AlexNet (CNN), MobileNet-LSTM	DL, HYBRID	CNN, HYBRID
Hossain et al. (2023)[44]	Bangladesh	AlexNet	DL	DCNN
Kanawade (2023)[45]	India	DenseNet-121	DL	DCNN
Nugroho et al. (2023)[46]	Indonesia	Inception-V3, DenseNet-201	DL	DCNN
Olayah et al. (2023)[16]	Saudi Arabia	AlexNet-ANN, AlexNet-GoogLeNet-VGG16-ANN, AlexNet-GoogLeNet-RF, GoogLeNet-VGG16-RF, AlexNet-VGG16-RF, GoogLeNet-ANN, AlexNet-RF, GoogLeNet-RF, VGG16-RF, AlexNet-GoogLeNet-ANN, GoogLeNet-VGG16-ANN, AlexNet-VGG16-ANN	HYBRID	HYBRID
Orhan (2023)[47]	Türkiye	CNN (MobileNet)	DL	CNN
Shehzad and Zhenhua (2023)[48]	China	Swin Transformer – EfficientNetV2S	HYBRID	HYBRID
Shen (2023)[49]	China	HarDNet (CNN)	DL	CNN
Tasar (2023)[50]	Türkiye	DenseNet121, VGG16, ResNet50, MobileNet	DL	DCNN
Wei (2023)[51]	China	CNN	DL	CNN
Wu (2023)[52]	Taiwan	MEL-HSNet – ResNet50	HYBRID	HYBRID

TABLE 1 (Continued)

Authors	Country	Used AI models	AI models (Scenario I)	AI models (Scenario II)
Ali (2024)[53]	Switzerland	DenseNet201	DL	DCNN
Faizi (2024)[54]	Pakistan	SVM	ML	SVM
Saha (2024)[55]	Bangladesh	ViT (CNN)	DL	CNN

Abbreviations: ANN, artificial neural network; CNN, convolutional neural network; DCNN, deep convolutional neural network; DL, deep learning; KNN, K-nearest neighbours; LSTM, long short-term memory; MEL-HSNet, multi-environmental learning–hyper spectral network; mi-SVM multi-instance-support vector machine; ML, machine learning; NasNet Large ACO, neural architecture search network large ant colony optimization; NasNet Large PSO, neural architecture search network large particle swarm optimization; RDCNN, residual deep convolution neural network; RF, random forest; SVM, support vector machine; VGG, visual geometry group; ViT, vision transformers; YOLO you only look once.

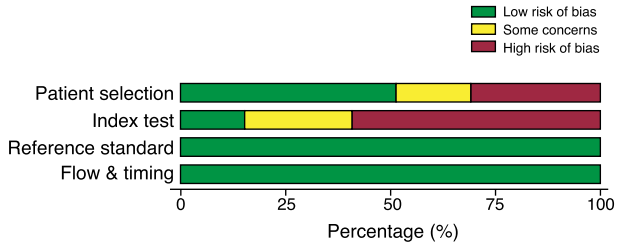


FIGURE 2 Summary of the traffic light graph reviewing authors' judgements about each domain for each included study in the quantitative analysis ($n = 78$) using the Quadas-2 tool.

Quality assessment

According to the QUADAS-2 tool, studies where at least two evaluations (low, some and high concern) differed across the four main domains (patient selection, index test, reference standard, flow and timing) include Kassani et al. (2019), Guo et al. (2020), Tiwari (2020), Hosny (2019), Damarla et al. (2022), and Nie et al. (2022). In these studies, the domain of 'patient selection' was found to have the highest level of concern, whereas the 'index test' domain was associated with a moderate level of concern, and other domains were rated as low concern. In the study by Afza et al. (2019), the 'patient selection' domain was also evaluated with a high level of concern. Unver (2019), Modooei (2019), Tang (2019), Banerjee et al. (2021), Bansal et al. (2021), Thanh (2021) and Nugroho et al. (2023) evaluated both the 'patient selection' and 'index test' domains with a high level of concern. In the studies by Anand et al. (2021), Aljohani and Turki (2022), Mahum et al. (2022), Qureshi et al. (2022), Alenezi et al. (2023), Kanawade (2023), Orhan (2023), Shehzad & Zhenhua (2023), Shen (2023), Wu (2023), Faizi (2024) and Saha (2024), the 'index test' was associated with a high level of concern, while other domains were found to have a low level of concern. Wang et al. (2022), Yılmaz et al. (2022), Akbulut et al. (2023), Olayah et al. (2023), Wei et al. (2023) and Ali et al. (2024) found the 'index test' domain to have some concerns, while other domains were rated with a low level of concern. The studies by Araujo et al. (2022) and Taşar (2023) reported low levels of concern across all domains (Figure 2 and Figure S2).

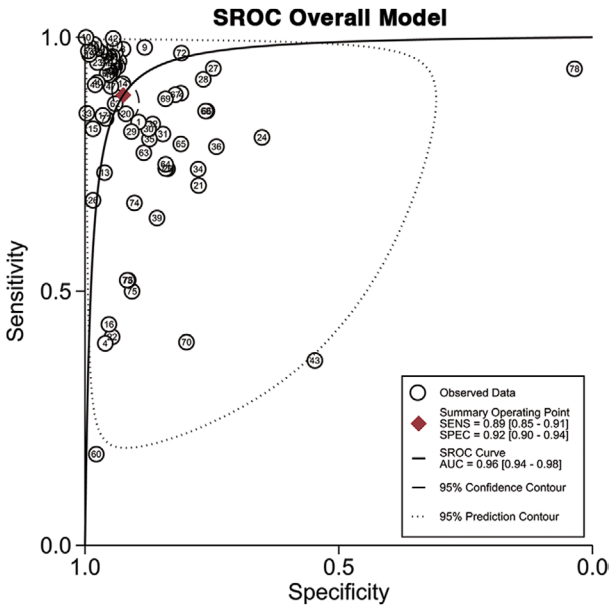


FIGURE 3 Summary receiver operator characteristics curves summary with confidence contours and prediction zones around the average sensitivity and specificity point. AUC, area under the curve; SENS, sensitivity; SPEC, specificity; SROC, summary receiver operator characteristics curves.

Diagnostic accuracy

The studies with the highest sensitivity were those conducted by Saha in 2024 and Alenezi et al. in 2023, while those performed by Baigh et al. in 2023 had the lowest sensitivity. Of the studies reviewed, 13 demonstrated a sensitivity performance below 70%, whereas 65 exhibited a sensitivity performance exceeding 70%. Another study with the highest specificity was conducted by Saha in 2024 and Damarla in 2022. Shen in 2023 had the lowest specificity. Only four studies reported a specificity below 70%, whereas 76 studies demonstrated a specificity exceeding 70% (Figure S3).

Overall model

The pooled area under the curve (AUC) estimated using SROC for the overall AI in this study was 0.96 [95% CI:

0.94–0.98]. The SROC curve plot is presented (Figure 3). According to the results, the pooled sensitivity was 0.89 [95% CI: 0.85–0.91], and the pooled specificity was 0.92 [95% CI: 0.90–0.94]. A Fagan nomogram was developed to predict how AI outcomes increase the risk of a patient having skin cancer (Figure S4). With a pre-test probability of 20%, the probability of a positive test result was 78%, and the probability of a negative test result was 2%. The post-test probability of a negative result is 2%.

Subgroup analysis

In Scenario-I, studies included 54 studies with DL models, 18 with HYBRID models and 6 with ML models. In the subgroup analysis, the AUC of the models for the first scenario was found as follows: HYBRID models 0.98 [95% CI: 0.96–0.99], DL models 0.95 [95% CI: 0.93–0.97] and ML models 0.95 [95% CI: 0.92–0.96] (Figure 4). Sensitivities were found as follows: HYBRID models 0.9 4 [95% CI: 0.89–0.97], DL models 0.86 [95% CI: 0.81–0.89] and ML models 0.86 [95% CI: 0.69–0.94]. The HYBRID model exhibited the highest AUC and sensitivity. For specificity, the values were DL model—0.93 [95% CI: 0.90–0.95], HYBRID model—0.95 [95% CI: 0.93–0.96] and ML model—0.91 [95% CI: 0.77–0.97]. The HYBRID model demonstrated the highest specificity.

In Scenario-II, the studies were included: 14 from CNN models, 40 from DCNN models, 18 from HYBRID models and six from SVM models. In the subgroup analyses, the AUC of the models in the first–second scenario was found as follows: HYBRID models 0.98 [95% CI: 0.96–0.99], CNN models 0.95 [95% CI: 0.92–0.96], DCNN models 0.96 [95% CI: 0.94–0.98] and SVM models 0.95 [95% CI: 0.92–0.96] (Figure 5). Sensitivities were as follows: HYBRID model—0.94 [95% CI: 0.89–0.97], CNN model—0.84 [95% CI: 0.70–0.92], DCNN model—0.89 [95% CI: 0.84–0.92] and SVM model—0.86 [95% CI: 0.69–0.94]. The highest AUC and sensitivity were observed in the HYBRID model. For specificity, the values were CNN model—0.92 [95% CI: 0.83–0.97], DCNN model—0.93 [95% CI: 0.89–0.95], HYBRID model—0.95 [95% CI: 0.93–0.96] and SVM model—0.91 [95% CI: 0.77–0.97]. The highest specificity was found in the HYBRID model.

Publication bias

It was found that studies with higher sample sizes were included more in the meta-analysis. In our study, we have observed that the studies have large sample sizes and are homogeneously distributed.

DISCUSSION

Integrating AI into dermatology markedly enhances the detection of skin cancer, notably melanoma. At this juncture, dermatologists are essential not only for identifying malignancies and excising suspicious lesions but also for effectively communicating results to patients.[56] The timeline from diagnosis to intervention is vital for optimizing patient management. Moreover, the efficacy of AI in dermatology is considerably impacted by image quality and the clarity of annotations, which influence diagnostic performance and interpretation.[57] Therefore, addressing these factors is crucial for advancing AI applications in skin cancer detection and improving clinical outcomes.[57] Furthermore, AI integration in dermatology is supported by evidence. A 2023 systematic review of 40 studies underscored AI models' potential in non-invasive melanoma imaging, demonstrating comparable or superior performance to dermatologists.[58] Recent studies further indicate that AI methods excel in classifying skin lesions, with average AUC values reaching as high as 0.96.[59] In anticipation of previous studies, our meta-analysis comprehensively evaluates the performance of ML, DL and HYBRID approaches for melanoma detection. The findings demonstrate that AI methods excel in classification tasks, achieving an impressive average AUC of 0.96 [95% CI: 0.94–0.98] using dermoscopic image features. This high AUC suggests a substantial capability of AI applications in predicting melanoma risk when utilizing dermoscopic images, demonstrating their potential as diagnostic aids in clinical settings. In a meta-analysis conducted by Nelson et al., various CAD models were evaluated, revealing an average sensitivity of 0.74 and specificity of 0.84 for melanoma detection across multiple studies.[60] However, this study lacked a detailed analysis of the predictors and classification models. Contrastingly, findings reported in other high-quality literature provide a comprehensive overview, indicating that advanced DL models can achieve higher average sensitivity and specificity, with metrics such as a sensitivity of 0.89 and specificity of 0.92 for image-based models being observed.[59,61] Furthermore, in subgroup analyses, Suli Li found the highest sensitivity of 0.94 [95% CI: 0.87–0.97] with HYBRIDs, while SVM models achieved the highest specificity with a result of 0.94 [95% CI: **0**.89–0.97].[9] In our study, subgroup analyses showed that the highest sensitivity of 0.94 [95% CI: 0.89–0.97] and specificity of 0.95 [95% CI: 0.93–0.96] were achieved by HYBRID. Our study found that HYBRID modelling yielded the highest performance. Ahmed noted increased heterogeneity due to combining different algorithms and potentially unspecified terms.[62] Similarly, our study found that including large and small sample sizes and combining various models contributed to high heterogeneity.

These studies reflect an acknowledgment of AI's potential to augment, rather than replace, the role of dermatologists, thereby improving early detection outcomes through more efficient triaging of lesions.[63] To adopt AI tools effectively, continuous efforts should focus on improving their accuracy and overcoming inherent biases, with a critical eye toward optimizing training datasets that reflect diverse populations.[63,64] Consequently, AI systems, particularly those leveraging DL and HYBRID models, present a viable avenue

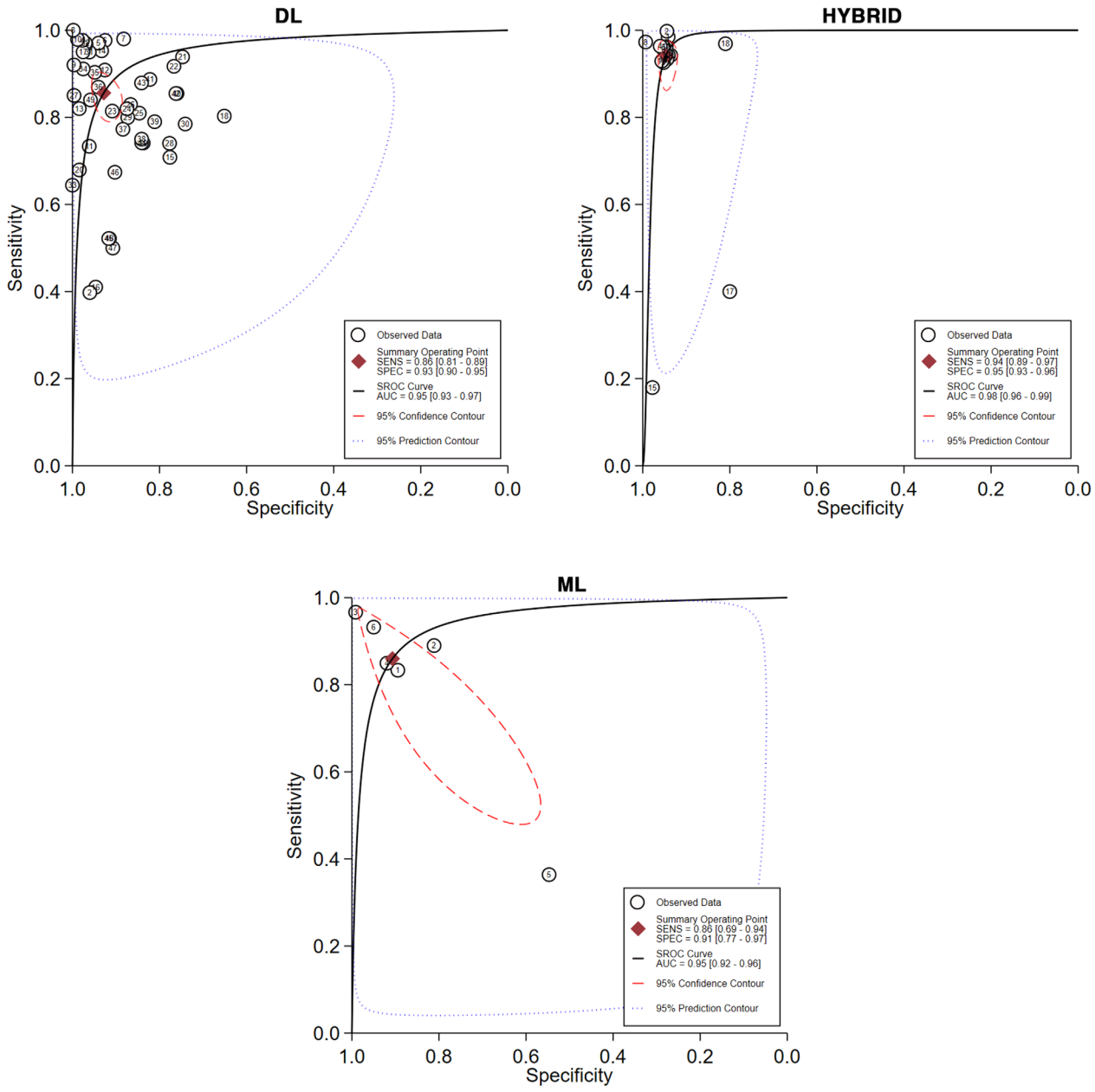


FIGURE 4 ROC curves comparing different AI models for subgroup prediction of AI models 1 prediction. The prediction in melanoma was associated with a pooled AUC of 0.95 (95% CI: 0.93–0.97) for DL algorithms, a pooled AUC of 0.98 (95% CI: 0.96–0.99) for hybrid models and pooled AUC of 0.95 (95% CI: 0.92–0.96) for ML algorithms. AUC, area under the curve; SENS, sensitivity; SPEC, specificity; SROC, summary receiver operator characteristics curves.

for enhancing the early detection of melanoma while providing essential support to dermatologists in their clinical decision-making processes.[65,66]

LIMITATIONS

One of the main limitations of our study is that ML and SVM models are less preferred in the current literature. For this reason, the sample pool is more affected by the performance of the studies included. Moreover, the fact that the performance of the included studies directly affects the effect size has led to similar performance of models such as ML and SVM compared to other methods such as CNN and DL. This has made it difficult to examine the effect of model diversity on the results in more detail.

CONCLUSIONS

Recent research trends suggest an anticipated rise in HYBRID model use. This development promises to provide valuable insights for dermatologists, as models that play a critical role in skin cancer screening, based on noteworthy

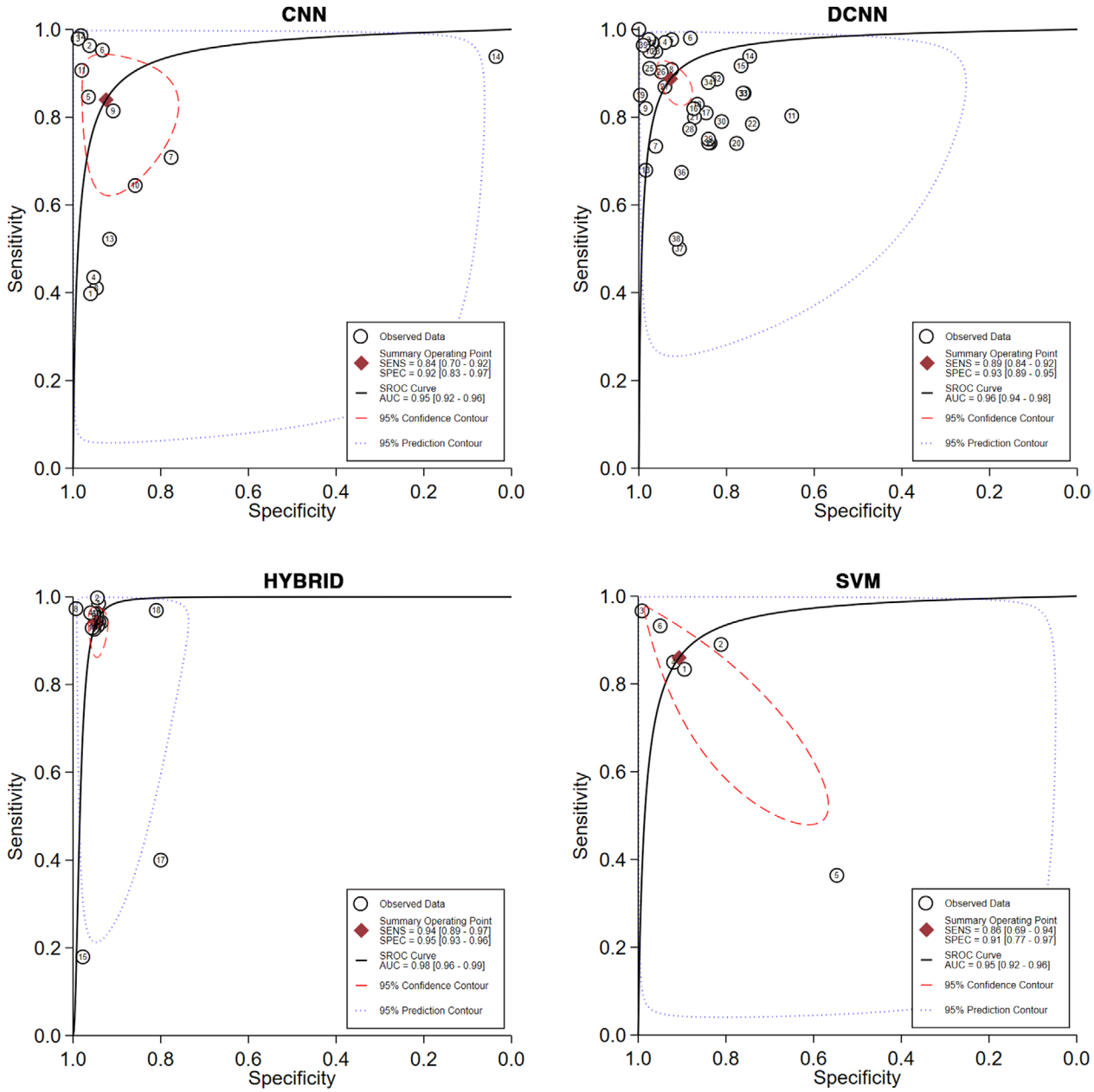


FIGURE 5 ROC curves comparing different AI models for subgroup prediction of AI models 2 prediction. The prediction in melanoma was associated with a pooled AUC of 0.95 (95% CI: 0.92–0.96) for CNN algorithms, a pooled AUC of 0.96 (95% CI 0.94–0.98) for DCNN algorithms, a pooled AUC of 0.98 (95% CI: 0.96–0.99) for hybrid models and pooled AUC of 0.95 (95% CI: 0.92–0.96) for SVM algorithms. AUC, area under the curve; SENS, sensitivity; SPEC, specificity; SROC, summary receiver operator characteristics curves.

features selected from lesion images in the existing database, are on the horizon. We estimate that HYBRID models will be used more, and experimental studies evaluating the performance of these models will increase.

AUTHOR CONTRIBUTIONS

Gözde Ertürk Zararsız: supervision, formal analysis, conceptualisation, writing—original draft, writing—review and editing. **Serra Ilayda Yerlitaş Taştan:** investigation, formal analysis, conceptualisation, writing—original draft, writing—review and editing. **Elif Çelik Gürbulak:** investigation, formal analysis, conceptualization. **Aleyna Erakcaoğlu:** investigation, formal analysis, writing—original draft, writing—review and editing. **Selen Yılmaz Işıkhan:** investigation, writing—review and editing. **Abdullah Demirbaş:** investigation, writing—review and editing, **Ragıp Ertaş:** investigation, writing—review and editing, **İrem Eroğlu:** investigation, writing—review and editing, **Selçuk Korkmaz:** investigation, writing—review and editing, **Ömer Faruk Elmas:** investigation, writing—review and editing, **Gökmen Zararsız:** supervision, writing—review and editing.

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CONFLICT OF INTEREST STATEMENT
Gözde Ertürk Zararsız, Serra Ilayda Yerlitaş Taştan, Elif Çelik Gürbulak, Aleyna Erakcaoğlu, Selen Yılmaz Işıkhan, Abdullah Demirbaş, Ragıp Ertaş, İrem Eroğlu, Selçuk Korkmaz, Ömer Faruk Elmas and Gökmen Zararsız declare that they have no conflicts of interest that might be relevant to the contents of this manuscript.

DATA AVAILABILITY STATEMENT
The datasets generated and/or analysed during the current study are available from the corresponding author on reasonable request.

ETHICAL APPROVAL
Not applicable.

ETHICS STATEMENT
Not applicable.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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